Do Now · retrieval

Lesson 109 · CPU Scheduling: FCFS, SJF and SRT

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Name the routine executed when an interrupt is accepted.

Q2

An OR gate has inputs 0 and 1. What is the output?

Q3

What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q4

What is the time complexity of sequential search in the worst case?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

FCFS — First come, first served: non-preemptive queue in arrival order.

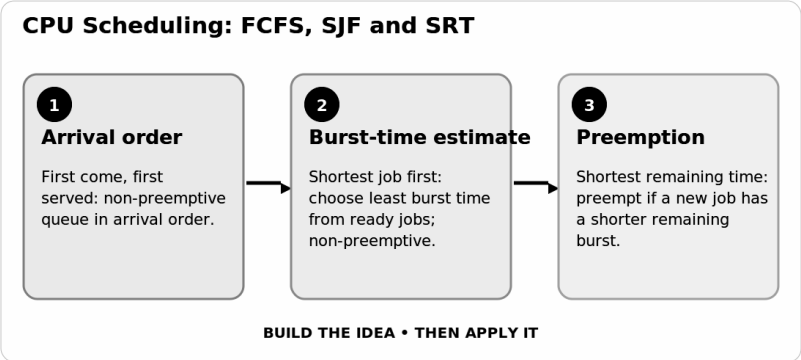
Core idea 2

SJF — Shortest job first: choose least burst time from ready jobs; non-preemptive.

Core idea 3

SRT — Shortest remaining time: preempt if a new job has a shorter remaining burst.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

FCFS — Jobs run in first-arrival order.

SJF — Selects the shortest estimated ready job.

SRT — Preemptive shortest remaining time.

waiting time — Time spent ready but not executing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

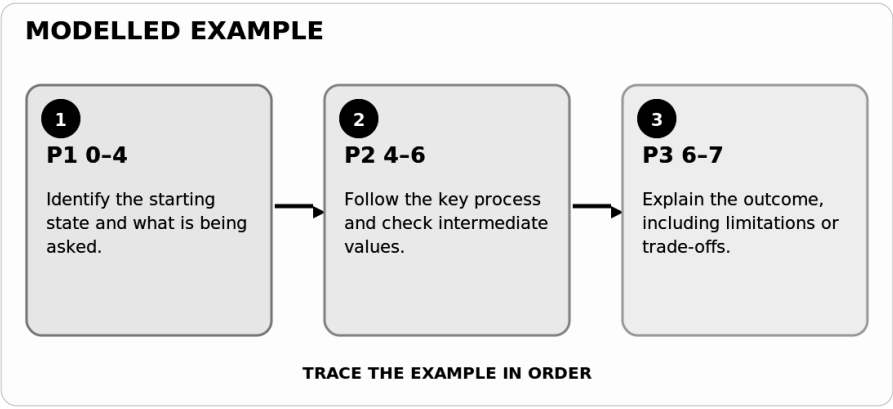
Problem

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

Follow the worked method

1. Model answer — Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Distinguish non-preemptive and preemptive scheduling and give an example of each. [6]

Question 2

FCFS: A arrives 0/burst 3, B arrives 1/burst 2, C arrives 2/burst 4. Calculate each wait. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

SRT: A arrives 0/burst 5, B arrives 1/burst 2. Show the timeline and explain preemption. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

SRT never interrupts an executing process.

Use key vocabulary from memory

1. Explain FCFS in your own words.

2. Explain SJF in your own words.

3. Explain SRT in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.